

UNIVERSITY NAME

*Probability and Statistics — Final Exam**Year: 2019***Exercise 1:**

Let X be a random variable uniformly distributed on $[0, 1]$ with probability density function

$$f_X(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Define $Y = -\ln X$.

1. Find the probability density function of Y .
2. Compute $P(|Y - 1| > \frac{1}{10})$.

Answer Area

1. Probability density function of Y

Since $X \sim \mathcal{U}([0, 1])$, and $Y = -\ln X$. The transformation is strictly decreasing. The CDF method: For $y > 0$,

$$F_Y(y) = P(Y \leq y) = P(-\ln X \leq y) = P(X \geq e^{-y}) = 1 - F_X(e^{-y}).$$

Since $f_X(x) = 1$ for $x \in [0, 1]$, $F_X(x) = x$ for $0 \leq x \leq 1$. For $y > 0$, $e^{-y} \in (0, 1]$, so:

$$F_Y(y) = 1 - e^{-y}.$$

Differentiating:

$$f_Y(y) = \frac{d}{dy}(1 - e^{-y}) = e^{-y}, \quad y > 0.$$

Thus,

$$f_Y(y) = e^{-y} \mathbf{1}_{(0, \infty)}(y).$$

2. Compute $P(|Y - 1| > \frac{1}{10})$

$$P\left(|Y - 1| > \frac{1}{10}\right) = 1 - P\left(|Y - 1| \leq \frac{1}{10}\right) = 1 - P(0.9 \leq Y \leq 1.1).$$

$$P(0.9 \leq Y \leq 1.1) = \int_{0.9}^{1.1} e^{-y} dy = [-e^{-y}]_{0.9}^{1.1} = e^{-0.9} - e^{-1.1}.$$

Thus,

$$P\left(|Y - 1| > \frac{1}{10}\right) = 1 - (e^{-0.9} - e^{-1.1}) = 1 - e^{-0.9} + e^{-1.1}.$$

$$1 - e^{-0.9} + e^{-1.1}.$$

Exercise 2:

A player starts a video game and plays several matches. It is assumed that the probability of winning the first match is 0.1:

- If they win a match, the probability of winning the next one is 0.8.
- If they lose a match, the probability of winning the next one is 0.6.

Let G_n be the event “the player wins the n -th match” and let $p_n = P(G_n)$.

1. Calculate p_2 .
2. Given that the player won the second match, calculate the probability that they lost the first one.
3. Find the probability that the player wins at least one match in the first three matches.
4. Show that for all natural numbers $n \geq 1$, we have

$$p_{n+1} = \frac{1}{5}p_n + \frac{3}{5}.$$

5. Assume the sequence $(p_n)_{n \geq 1}$ converges to a limit ℓ . Determine ℓ and show that the sequence $(p_n - \ell)_{n \geq 1}$ is geometric.
6. Deduce an expression for p_n as a function of n .

Answer Area

Given: $P(G_1) = p_1 = 0.1$, and

$$P(G_{n+1} | G_n) = 0.8, \quad P(G_{n+1} | G_n^c) = 0.6.$$

1. Calculate p_2

$$p_2 = P(G_2) = P(G_2 | G_1)P(G_1) + P(G_2 | G_1^c)P(G_1^c) = 0.8 \cdot 0.1 + 0.6 \cdot 0.9 = 0.62.$$

2. Given G_2 , probability they lost the first one

$$P(G_1^c | G_2) = \frac{P(G_2 | G_1^c)P(G_1^c)}{P(G_2)} = \frac{0.6 \cdot 0.9}{0.62} = \frac{0.54}{0.62} = \frac{27}{31}.$$

3. Probability that the player wins at least one match in the first three matches. Easier to compute complement: loses all three matches. Let $L_i = G_i^c$. Then:

$$P(L_1) = 0.9, \quad P(L_2 | L_1) = 1 - 0.6 = 0.4,$$

$$P(L_3 | L_2) = 1 - 0.6 = 0.4 \quad (\text{since after loss, win prob } 0.6)$$

But careful: the loss-loss transition probability is actually $P(L_{n+1} | L_n) = 1 - 0.6 = 0.4$, yes.

So:

$$P(L_1 \cap L_2 \cap L_3) = P(L_1)P(L_2 | L_1)P(L_3 | L_2) = 0.9 \cdot 0.4 \cdot 0.4 = 0.144.$$

Thus:

$$P(\text{at least one win}) = 1 - 0.144 = 0.856.$$

4. Show that $p_{n+1} = \frac{1}{5}p_n + \frac{3}{5}$

$$\begin{aligned} p_{n+1} &= P(G_{n+1}) = P(G_{n+1} \mid G_n)P(G_n) + P(G_{n+1} \mid G_n^c)P(G_n^c) \\ &= 0.8p_n + 0.6(1 - p_n) = 0.8p_n + 0.6 - 0.6p_n = 0.2p_n + 0.6 = \frac{1}{5}p_n + \frac{3}{5}. \end{aligned}$$

5. Limit ℓ and show $(p_n - \ell)$ geometric If limit ℓ exists, then from recurrence:
 $\ell = \frac{1}{5}\ell + \frac{3}{5} \Rightarrow \ell - \frac{1}{5}\ell = \frac{3}{5} \Rightarrow \frac{4}{5}\ell = \frac{3}{5} \Rightarrow \ell = \frac{3}{4}$. Now consider $d_n = p_n - \ell = p_n - \frac{3}{4}$.
 Then:

$$d_{n+1} = p_{n+1} - \frac{3}{4} = \left(\frac{1}{5}p_n + \frac{3}{5}\right) - \frac{3}{4} = \frac{1}{5}p_n - \frac{3}{20}.$$

But note $\frac{3}{20} = \frac{1}{5} \cdot \frac{3}{4}$, so:

$$d_{n+1} = \frac{1}{5}p_n - \frac{1}{5} \cdot \frac{3}{4} = \frac{1}{5} \left(p_n - \frac{3}{4}\right) = \frac{1}{5}d_n.$$

Thus (d_n) is geometric with ratio $\frac{1}{5}$.

$$\ell = \frac{3}{4}, \quad p_n - \ell \text{ is geometric with ratio } \frac{1}{5}.$$

6. Expression for p_n Since $d_1 = p_1 - \frac{3}{4} = 0.1 - 0.75 = -0.65 = -\frac{13}{20}$, then:

$$d_n = d_1 \left(\frac{1}{5}\right)^{n-1} = -\frac{13}{20} \cdot \left(\frac{1}{5}\right)^{n-1}.$$

Thus:

$$p_n = \ell + d_n = \frac{3}{4} - \frac{13}{20} \cdot \left(\frac{1}{5}\right)^{n-1}.$$